TRAN DINH PHU

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) *May 2028 (Expected)*
Ph.D. in Computer Science
GPA: 3.90/4.3 (Current GPA)
Advisor: Prof. Dr. Daeyoung Kim

Korea Advanced Institute of Science and Technology (KAIST) *February 2025*
M.S. in Computer Science
GPA: 3.94/4.3
Thesis: Channel-Partitioned Windowed Attention And Frequency Learning for Single Image Super-Resolution
Advisor: Prof. Dr. Daeyoung Kim

Hanoi University of Science and Technology (HUST) *June 2022*
B.E. in Control Engineering and Automation
GPA: 3.66/4.0 (*Top 1%*)
Thesis: Trans2Unet: Neural fusion for nuclei semantic segmentation (9.5/10 - Best Presentation Award)
Advisors: Assoc. Prof. Dr. Thi-Thao Tran and Assoc. Prof. Dr. Van-Truong Pham

Relevant Coursework: Computer Vision | Deep Learning | Advanced Deep Learning | Artificial Intelligence and Machine Learning | Unsupervised Learning and Generative Models | Audio-Visuali Language Multi-Modal AI | Computational Linguistics

EXPERIENCE & RESEARCH INTERESTS

Image and Video Restoration/Enhancement
Image and video super-resolution, restoration, image enhancement, and low-level vision. Research interests include efficient restoration architectures, diffusion-based generative restoration, perceptual–fidelity trade-offs, and foundation models for image and video enhancement.

Multimodal Learning and Reasoning
Audio-visual question answering, multimodal reasoning, and agentic AI systems. Research interests include reliable multimodal learning, parameter-efficient adaptation of foundation models, cross-modal representation learning, and scalable reasoning over long-horizon multimodal content.

Optimization and Model Generalization
Robust optimization, model generalization, and learning under limited data and computational budgets. Particular interests include Sharpness-Aware Minimization (SAM) and regularization techniques for improving robustness, fairness, and training stability.

Emerging Research Directions
Open to exploring new research topics and emerging problems in machine learning and multimodal AI. Recently interested in post-training multimodal large language models (MLLMs), with a particular focus on improving their reasoning capabilities, especially for complex visual and multimodal tasks.

PUBLICATIONS

EnComp: Lightweight Encoder-Only Context Compression for Retrieval-Augmented Question Answering
Thao Do, **Dinh Phu Tran**, An Vo, Seon Kwon Kim, Daeyoung Kim
AACL 2026 (Oral), Hengqin, China

FreqOrtho-SR: Frequency-Guided Orthogonal Expert Learning for Real-World Image Super-Resolution
Minh Son Hoang*, **Dinh Phu Tran***, Quyen Nguyen Duc, Dam Hoang Phuong, Daeyoung Kim
ECCV 2026, Malmö, Sweden

MedCAGD: Context-Aware Gated Decoder for Robust Medical Image Segmentation
Saad Wazir, Patrick Vibild, **Dinh Phu Tran**, Seongah Kim, Daeyoung Kim
ECCV 2026, Malmö, Sweden

Efficient Encoder-Only Context Compression via Marginal Contribution Scoring
Thao Do, **Dinh Phu Tran**, An Vo, Seon Kwon Kim, Daeyoung Kim
ICML 2026 Workshop, Seoul, South Korea

SAT: Selective Aggregation Transformer for Image Super-Resolution
Dinh Phu Tran, Thao Do, Saad Wazir, Seongah Kim, Seon Kwon Kim, Daeyoung Kim
CVPR 2026 (Finding Track), Denver, USA

VSRM: A Robust Mamba-Based Framework for Video Super-Resolution
Dinh Phu Tran, Dao Duy Hung, Daeyoung Kim
ICCV 2025, Hawaii, USA

Reference-Based Post-OCR Processing with LLM for Precise Diacritic Text in Historical Document Recognition
Thao Do, **Dinh Phu Tran**, An Vo, Daeyoung Kim
AAAI 2025 (Social Track), Philadelphia, Pennsylvania, USA

Channel-Partitioned Windowed Attention And Frequency Learning for Single Image Super-Resolution
Dinh Phu Tran, Dao Duy Hung, Daeyoung Kim
BMVC 2024, Glasgow, UK

PAPER UNDER REVIEW

Knowing When to Answer: Adaptive Confidence Refinement for Reliable Audio-Visual Question Answering
Dinh Phu Tran, Jihoon Jeong, Saad Wazir, Seongah Kim, Thao Do, Cem Subakan, Daeyoung Kim

FalOTS: Orthogonal Target-Aware SAM for Federated Active Learning Under Global Class Imbalance
Dinh Phu Tran, Seon Kwon Kim, Thao Do, Dam Hoang Phuong, Minh Son Hoang, Saad Wazir, Seungmin Yang, Daeyoung Kim

Frequency-Aware Spectral-Spatial Interaction Network for Efficient Image Super-Resolution
Dinh Phu Tran, Seungmin Yang, Thao Do, Daeyoung Kim

HONORS AND AWARDS

Huyndai Global Scholarship for Graduate Student (5 slots/admission cycle)	*2025-2029*
Kaist Scholarship for PhD Program (Full tutions and fees)	*2025-2029*
Kaist Scholarship for Master Program (Full tutions and fees)	*2023-2025*
Best Bachelor Thesis Award	*09/2022*
Scholarship Encouraging Study (Level A - Highest Level)	*2020-2022*
Nitori International Scholarship	*2019-2020*
Graduated as valedictorian from high school	*09/2018*

SKILLS

Programming Language	Python (Framework - Pytorch), Bash, C/C++
Deep Learning	Neural Nets (CNN, Transformer, Mamba), Generative Models, MLLMs
Version Control Systems	Git
Language	English, Vietnamese (mother tongue)

ACADAMIC SERVICES

Reviewer for AAAI 2026-2027, CVPR 2026, ECCV 2026, ICML 2026, NeurIPS 2026